Interoperability Conformance Specification: spectralEncoding – Basic and Advanced

Contents

Foreword

This document has been prepared following the ICC Intellectual Property Policy. This policy is based on the ITU-T/ITU-R/ISO/IEC Guidelines for Implementation of the Common Patent Policy (23 April 2012), with interpretations and clarifications to make it specific to ICC. A Patent Statement and Licensing Declaration form is available.

ICC Interoperability Conformance Specifications, of which this document is an example, may be submitted to the competent ISO Technical Committee for consideration and development as an ISO document. If so, this foreword is to be replaced by the appropriate wording supplied by ISO.

Introduction

ISO 20677-1 defines specifications that provide a platform for defining extended (iccMAX) colour management profiles and systems for various colour workflow domains. It provides a platform for which domain specific specifications can be defined that make use of iccMAX extensions to the existing cross-platform profile format of ISO 15076-1. Thus there is greater flexibility for defining colour transforms and profile connection spaces to meet needs that cannot easily be met with ISO 15076-1. It is not envisioned that all colour management systems that use ISO 20677-1 will implement all the features or capabilities it specifies. Requirements specifying restrictions to iccMAX that apply to a particular workflow are defined in workflow domain specifications known as Interoperability Conformance Specifications, of which this document is one example. Additionally, for some domain specific workflows it is envisioned that workflows will connect both to profiles defined by ISO 20677-1 (iccMAX) and those defined by ISO 15076-1.

An Interoperability Conformance Specification (ICS) is approved and registered by the International Color Consortium (ICC). It defines minimum structural and operational requirements for writing and reading ICC profiles in order to address a specific problem and/or functionality that cannot readily be handled using the profile format defined by ISO 15076-1. An ICS document essentially defines restrictions to ISO 20677-1 for a specific use case.

Interoperability Conformance Specification: spectralEncoding – Basic and Advanced

1 Scope

All parts of this specification define scenario requirements and restrictions to profiles based on ISO 20677-1 for the purpose of connecting multi-spectral data with a spectrally-based Profile Connection Space (PCS).

The particular sub-set of the tags defined in ISO 20677-1 that are required to be present is defined, together with any optional tags that are permitted. The connections between profiles are described, and the processing elements that the CMM is required to support are identified.

This ICS defines transforms with basic limited processing element support, thus enabling spectral imaging workflows with fewer implementation requirements, as well as transforms that utilize advanced or extended processing element support for more complicated modeling and colour management situations. The distinction between basic and advanced shall be identified by the subclass version in the profile. Some implementations may choose to only implement support for basic (subclass 1) profiles conforming to this specification.

This ICS supports both full spectral encoding, in which each device channel corresponds to a spectral PCS band, and abridged (multi-spectral) encoding, in which a smaller number of device channels are used to reconstruct the spectral PCS. Profiles and workflows conforming to this ICS use only iccMAX (ISO 20677-1) profiles.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 20677-1:20##, *Image technology colour management — Extensions to architecture, profile format, and data structure: iccMAX*

NOTE: The most recent version of the iccMAX specification is available on the ICC web site [2].

3 Terms and definitions

All terms and definitions relevant to this document are provided in ISO 20677.

3.1 observing conditions

the observer's colour matching functions and illuminant associated with the spectralViewingConditoinsTag of the Profile Connection Condition (PCC) used to connect the profile to the PCS.

4 Use case

4.1 Domain

Profiles and workflows conforming to this ICS shall apply to the domain of General (i.e. the domain of application is unspecified).

4.2 Intended use

The intended use of this specification is to define workflows in which multi-spectral reflectance data are converted to or from a spectral reflectance PCS. Subsequent conversion to colorimetry can be achieved by applying the illuminant and observer specified in the spectralViewingConditionsTag.

4.3 Restrictions

ISO 20677-1 provides full details of the requirements for iccMAX profiles. This document defines a set of restrictions which apply to profiles created for the specific use case described above. Such restrictions include the sub-set of tags from ISO 20677-1 which are permitted in profiles conforming to this document.

5 Workflow

5.1 Profile sub-classes

A supporting CMM shall support the sub-classes of the profile classes identified in Table 1 for conformance with this ICS. All profile classes are defined in ISO 20677-1.

Table 1. Sub-classes of profile classes defined by this ICS

Profile	Profile class	Sub-class Identifier	Class signature	Sub-class signature
R	colorSpace	spectralEncoding	'spac'	'sref'
R	Input Device	spectralEncoding	'scnr'	'sref'

5.2 Connection Scenarios

5.2.1 General

A 'sref' spectralEncoding profile connects *x* input channels (where *x* corresponds to the number of channels of the Data colour space field of the header) to a spectral Profile Connection Space containing *n* spectral channels (where *n* corresponds to the number of spectral PCS channels defined in the header). It can be used as either a source profile, a destination profile, or as a PCC override.

5.2.2 Profiles for ICS workflow scenarios

Table 2 provides overview information about additional profiles that are used to describe several profile connection scenarios conforming to this ICS.

Table 2. Additional profiles referenced in workflow scenarios defined by this ICS

Profile	Description
C	Profile conforming to ISO 20677-1 or ISO 15076-1 containing (a) colorimetric transform(s) with CMM configured to use a colorimetric transform type
S	Profile conforming to ISO 20677-1 containing (a) spectral reflectance transform(s) with the CMM configured to use a spectral transform type
P	Profile conforming to ISO 20677-1 containing Profile Connection Condition tags (spectralViewingConditionsTag, customToStandardPccTag, standardToCustomPccTag)
1	Profile conforming to ISO 20677-1 utilizing Profile Connection Conditions to go into or out of the PCS
2	Profile conforming to ISO 20677-1 or ISO 15076-1

5.2.3 Workflow Scenarios

5.2.3.1 General

The following sections document scenarios that conform to this ICS.

5.2.3.2 Connecting a spectralEncoding profile as source to a destination profile with a colorimetric PCS

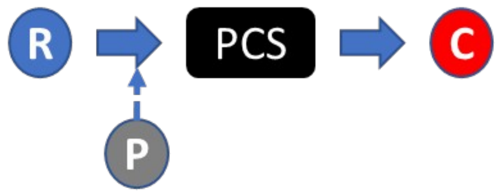
Profile Sequence	Profile Setup
R → PCS → C	R — Rendering Intent: Any Transform Type: Spectral PCC Override: None C — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None

In this scenario a profile conforming to this ICS is used as a source profile with a spectral PCS connected to an arbitrary profile that is set up to use colorimetric PCS. The transform type for profile R is "spectral" indicating that the transform from a DToBxTag from profile R is used to transform device values to a spectral reflectance PCS.

The observing conditions of profile R are used to convert from spectral reflectance to tristimulus XYZ colorimetry. If the observing conditions from profile R do not match the observing conditions used for profile C then the transform from the CustomToStandardTag in profile R is applied.

The resulting colorimetry is then transformed and applied as needed according to the rules (ICS or specification) associated with profile C.

5.2.3.3 Connecting a spectralEncoding profile as source using a PCC override to a destination profile with a colorimetric PCS

Profile Sequence	Profile Setup
	R — Rendering Intent: Any Transform Type: Spectral PCC Override: P C — Rendering Intent: Any Transform Type: Colorimetric PCC Override: None

In this scenario a profile conforming to this ICS is used as a source profile with a spectral PCS connected to an arbitrary profile that is set up to use colorimetric PCS. The transform type for profile R is "spectral" indicating that the transform from a DToBxTag from profile R is used to transform device values to a spectral reflectance PCS.

Since a profile override is specified, the observing conditions of profile P are used to convert from spectral reflectance to tristimulus XYZ colorimetry. If the observing conditions from profile P do not match the observing conditions used for profile C then the transform from the CustomToStandardTag in profile P is applied.

The resulting colorimetry is then transformed and applied as needed according to the rules (ICS or specification) associated with profile C.

5.2.3.4 Connecting a spectralEncoding profile as source to a destination profile with a spectral PCS

Profile Sequence	Profile Setup
	R — Rendering Intent: Any Transform Type: Spectral PCC Override: None S — Rendering Intent: Any Transform Type: Spectral PCC Override: None

In this scenario a profile conforming to this ICS is used as a source profile with a spectral PCS connected to an arbitrary profile that is set up to a spectral PCS. The transform type for profile R is "spectral" indicating that the transform from a DToBxTag from profile R is used to transform device values to a spectral reflectance PCS.

The resulting spectral PCS data is then transformed and applied as needed according to the rules (ICS or specification) associated with profile S.

Note: Spectral PCS sampling, range adjustments, and/or spectral PCS type adjustments are performed as needed according to Annex A of ISO 20677:2019.

5.2.3.5 Connecting source profile with a spectral PCS to a spectralEncoding profile as destination

Profile Sequence	Profile Setup
	S — Rendering Intent: Any Transform Type: Spectral PCC Override: None R — Rendering Intent: Any Transform Type: Spectral PCC Override: None

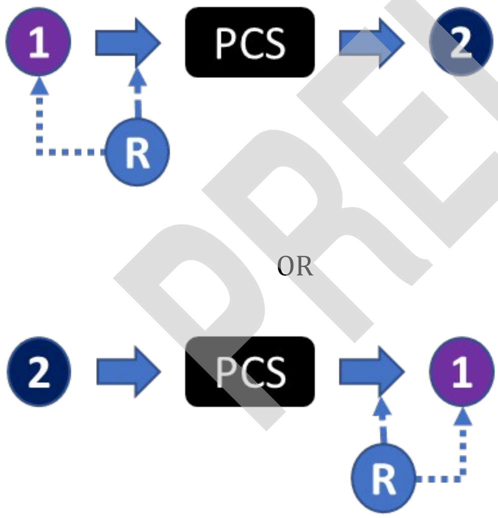
In this scenario an arbitrary profile that provided spectral PCS values is connected to a profile conforming to this ICS used as a destination profile.

Device values are first transformed to a spectral PCS according to the rules (ICS or specification) associated with profile S.

The resulting spectral PCS data is transformed to device values using the selected BToDxTag from profile R (since the transform type associated with profile R is 'spectral').

Note: Spectral PCS sampling, range adjustments, and/or spectral PCS type adjustments are performed as needed according to Annex A of ISO 20677:2019.

5.2.3.6 Using a spectralEncoding profile as a PCC override

Profile Sequence	Profile Setup
	1 — Rendering Intent: Any Transform Type: Any PCC Override: R 2 — Rendering Intent: Any Transform Type: Any PCC Override: None

In this scenario the profile connection conditions of a profile conforming to this specification are used as a PCC override of an arbitrary profile 1. The observing conditions may be used as part of the application of the transform or PCS conversion/adjustment for profile 1. Additionally, the transform from the standardToCustomPccTag or customToStandardPccTag of profile R may be applied as needed as part of the PCS conversion/adjustment for profile 1.

In this case all other transforms of profile R are ignored.

5.3 Other scenario implementation details

Application of abstract class profiles is also possible as part of additional PCS processing associated with the scenarios in section 5.2.

6 Sub-class Profile Requirements

6.1 General

Requirements for iccMAX profiles conforming to this ICS document are listed here. ICC v4 profiles shown in Section 5 above shall conform to ISO 15076-1.

6.2 Requirements

The requirements for the spectralEncoding sub-class of Colorspace class profiles are shown below.

The encoding of the profile header shall be as defined in ISO 20677-1, with the specific requirements shown in Table .

Table 3. Header requirements

Header field	Required content
Profile class	'spac' or 'scnr'
Profile subclass	'sref'
Profile subclass major version	1 for basic encoding, 2 for advanced encoding
Profile subclass minor version	0
Profile flags	0
Device attributes	0 or 1
Data colour space	'ncxxxx' or 'NCLR' where the channel count is the number of device channels in the encoding (equal to the number of spectral PCS channels for a full encoding, or fewer for an abridged multi-spectral encoding)
MCS	0
Colorimetric PCS	0
Spectral PCS	'rsxxxx' where xxxx is a hex value corresponding to the number of spectral channels in the PCS
Spectral PCS range	Start and end wavelengths of the spectral PCS encoded as float16 values
Bispectral PCS	0

Full details of the encoding of the header fields in Table 3 are given in ISO 20677-1.

6.2.1 Basic Encoding Required tags

Profiles with subclass major version of 1 shall contain the tags listed in Table 4.

Table 4. Required tags

Tag name	Signature	Required content
profileDescriptionTag	'desc'	multiLocalizedUnicodeType structure containing invariant and localizable versions of the profile description for display
copyrightTag	'cprt'	multiLocalizedUnicodeType containing copyright information for the profile
customToStandardPccTag	'c2sp'	multiProcessElementType containing a single 3x3 matrix element Note: this tag is not required if the spectralViewingConditions are for the D50 illuminant and standard 2-degree observer.
standardToCustomPccTag	's2cp'	multiProcessElementType containing a single 3x3 matrix element Note: this tag is not required if the spectralViewingConditions are for the D50 illuminant and standard 2-degree observer.
spectralWhitePointTag	'swpt'	float16NumberType containing an array of values defining the spectral reflectance or emission of the media white point
DToB3	'D2B3'	LutAToBType or multiProcessElementType using any combination of curveSetElement, matrixElement, CLUTElment, extendedCCLUTElement, and tintArrayElement
BToD3	'B2D3'	LutBToAType or multiProcessElementType using any combination of curveSetElement, matrixElement, CLUTElment, extendedCCLUTElement, and tintArrayElement
spectralViewingConditionsTag	'svcn'	Structure defining observer, illuminant and (optionally) surround

The encoding of the tags listed in Table 4 shall be as defined in ISO 20677-1.

6.2.2 Advanced Required tags

Profiles with subclass major version of 2 shall contain the same tags listed in Table 4 as defined in ISO 20677-1, with the expansion of the allowed processing elements for a multiProcessElementType in the DToB3 and BToD3 tags to all processing element types defined in ISO 20677-1, including the calculatorElement.

6.2.3 Example

The profiles Spec380_10_730-D50_2deg.icc (a full spectral encoding) and SixChanMsRef.icc (an abridged multi-spectral encoding) are encoded according to the requirements of this ICS document and are provided in the accompanying iccDEV-Testing ICS package (named SpectralEncoding) [3]. XML representations are also provided.

7 Conformance

A profile shall be considered to be in conformance with this ICS document if it meets the following conditions:

- The profile connects to the channels specified in section 5.
- The profile header includes the required content from Table 3.
- All required tags listed in Table 4 are present in the profile.
- The profile structure and all tags conform to ISO 20677-1.

A CMM shall be considered to be in conformance with this ICS if it meets the following conditions:

- The CMM is able to parse profiles that conform to this ICS
- The CMM supports and is capable of processing the channels specified in section 5 and any other profiles listed in Table 2 in the scenarios described in section 5.
- The CMM is able to process the tags listed in Table 4.
- When processing a profile conforming to this ICS, the CMM produces results that are a close approximation to those produced by the iccMAX Reference Implementation [3]

Bibliography

[1] ISO 20677-1:20##, *Image technology colour management — Extensions to architecture, profile format, and data structure*

[2] iccMAX http://www.color.org/iccmax/

[3] iccMAX Reference Implementation http://www.color.org/iccmax/index.xalter#reficcmax

8 Annex A (informative) - Worked example scenarios

This annex describes the worked example scenarios provided in the accompanying iccDEV-Testing ICS package named SpectralEncoding. Each entry is driven by a JSON configuration file in the package's config/ directory and demonstrates one of the connection scenarios defined in clause 5 applied to example spectralEncoding profiles. Full-encoding examples use the suffix 'a'; abridged (multi-spectral) examples use the suffix 'b'.

Table A.1. Worked example scenarios in the iccDEV-Testing ICS package

ID	Clause 5 scenario	iccDEV driver file	What is demonstrated
S1a	Scenario 1	iccApplyProfiles / se-S1a-refCowsToSrgbD50.json	Preview the full spectral cows image to sRGB under its native D50 observing conditions.
S1b	Scenario 1	iccApplyProfiles / se-S1b-ms6ToSrgb.json	Decode the abridged six-channel image to spectral, then preview to sRGB under its native D93 observing conditions.
S2a	Scenario 2	iccApplyProfiles / se-S2a-refCowsToSrgbA.json	Preview under Illuminant A via a colorimetric PCC override.
S2b	Scenario 2	iccApplyProfiles / se-S2b-refCowsToSrgbD65.json	Preview under D65 via a colorimetric PCC override (the metameric halves match).
S2c	Scenario 2	iccApplyProfiles / se-S2c-refCowsToSrgbD93.json	Preview under D93 via a colorimetric PCC override.
S2d	Scenario 2	iccApplyProfiles / se-S2d-refCowsToSrgbF11.json	Preview under F11 via a colorimetric PCC override.
S3a	Scenario 3	iccApplyProfiles / se-S3a-refCowsToSpec.json	Resample the 81-band spectral image into the 36-band full spectral encoding.
S3b	Scenario 3	iccApplyProfiles / se-S3b-ms6ToSpec.json	Reconstruct the full 36-band reflectance from the abridged six channels.
S4a	Scenario 4	iccApplyProfiles / se-S4a-specToRefCows.json	Re-encode the 36-band spectral image through the full encoding (round trip).

S4b	Scenario 4	iccApplyProfiles / se-S4b-specToMs6.json	Encode the 36-band spectral image into the abridged six-channel multi-spectral image.
S5	Scenario 5	iccApplyProfiles / se-S5-previewCowsMsPcc.json	Use the abridged profile purely as a D93 PCC override to re-render the cows.

8.1 Understanding the results

The smCows image is a metameric ("metacow") target: the head and tail halves of each cow share the same colour under CIE Illuminant D65 with the 1931 standard 2-degree observer, but differ under other illuminants. In the D65 preview (S2b) the two halves match; under Illuminant A (S2a), D93 (S2c) and F11 (S2d) they diverge. Because the encoding carries full spectral information, the previews reproduce this illuminant-dependent metamerism, which a single-illuminant colorimetric encoding cannot.

The abridged six-channel encoding preserves enough spectral information to reproduce the same behaviour: the abridged D93 preview (S1b) and the abridged PCC-override render (S5) closely match the full D93 preview (S2c), and the reflectance reconstructed from the six channels (S3b) closely matches the original 36-band spectral image (S3a).

Note: the abridged profile carries D93 observing conditions, so when it is used as a PCC override (S5) its standardToCustomPccTag and customToStandardPccTag transforms re-render the colorimetry under D93.